

Research Highlights

Counterfactual Evaluation of Causal Faithfulness in GINE Explanations for Molecular Toxicity Prediction

- We test whether GNN training induces causally faithful explanations.
- Counterfactual-sufficiency test (CCS) is validated on known-ground-truth probes.
- Standard ERM fails to align toxicophore attribution with decision boundaries.
- Causal regularization fails to improve alignment and degrades BACE consistency.
- Negative result replicated across 5 seeds, 3 datasets, and 3 metrics.